

Compromised Air Traffic Control Workstation

Case Reference: INC-2026-0624-0001

Examiner: niel · 2026-06-24 22:25 UTC

3

CRITICAL

4TOTAL
FINDINGS**9**

INDICATORS

01.0 EXECUTIVE SUMMARY

Incident Overview

Concise narrative of the incident — what happened, when, impact, and required actions.

3

CRITICAL SEVERITY

YESCOMPROMISE
CONFIRMED**19 min**

TIME TO C2

Phishing

ATTACK VECTOR

KEY FINDINGS

On 2025-07-22, an air traffic control workstation (TSO-ATC-CT412, user 'award') was compromised within 3 minutes of user login via a phishing-delivered trojan (epxplorer.exe) masquerading as explorer.exe. The malware established command-and-control to an attacker-operated Telegram Bot API behind Cloudflare infrastructure 19 minutes after execution. At the time of compromise, ATC_RADAR.exe was actively processing aviation radar data.

Required Action: Immediate network isolation of TSO-ATC-CT412 · Escalate to aviation safety officer · Potential ICAO Annex 17 reporting obligation

Event Reconstruction

Chronological sequence of events reconstructed from memory forensics artifacts. Each event is traced to a specific tool and finding reference.

TIME (UTC)	DATE	EVENT	TYPE	SOURCE TOOL	FINDING REF
18:50:46	2025-07-22	System boot	SYSTEM	volatility3 windows.info	—
18:50:50	2025-07-22	User 'award' logs in — explorer.exe (PID 5116) starts	USER	volatility3 windows.pslist	—
18:51:58	2025-07-22	Microsoft Edge establishes HTTPS connection (browsing activity begins)	USER	volatility3 windows.netscan	—
18:53:11	2025-07-22	MALWARE EXECUTION: epnlorer.exe (PID 9920) spawned by explorer.exe	MALWARE	volatility3 windows.pslist + MemProcFS	F-niel-001
19:12:33	2025-07-22	C2 ESTABLISHED: epnlorer.exe → 104.21.1.247:443 (Telegram Bot API)	C2	volatility3 windows.netscan	F-niel-002
19:15:05	2025-07-22	ATC_RADAR.exe active — malware coexisting with aviation operations software	PROCESS	volatility3 windows.pslist + MemProcFS	F-niel-004
19:18:53	2025-07-22	Memory dump captured — evidence acquisition point	SYSTEM	volatility3 windows.info	—

Type legend: Malware Network/C2 Process System User

Detailed Investigation Findings

Each finding includes the observation, interpretation, impact assessment, MITRE ATT&CK mapping, tool traceability, and remediation guidance.

F-NIEL-001

CRITICALHIGH

Malware Execution: epexplorer.exe Masquerading as explorer.exe

Observation: Process epexplorer.exe (PID 9920) was spawned by explorer.exe (PID 5116) at 2025-07-22 18:53:11 UTC, just 2 minutes 21 seconds after user login. The binary was executed from C:\Users\award\Downloads\epexplorer.exe.

Impact: The malware achieved code execution on an air traffic control workstation within 3 minutes of user login. The filename explicitly targets the Windows shell process (explorer.exe → epexplorer.exe, substituting 'l' with '1') to evade casual observation in Task Manager. This is a targeted attack against an aviation-critical system.

MITRE ATT&CK	TOOL	EVIDENCE	CROSS-VALIDATION
T1036.005, T1204.002	volatility3 2.7.0 + MemProcFS 5.17.8	EVID-001 · raw/vol3_pslist.csv	MemProcFS proc.txt confirmed identical process tree

F-NIEL-002

CRITICALHIGH

Command and Control via Fraudulent Telegram Bot API

Observation: epexplorer.exe established an outbound HTTPS connection to 104.21.1.247:443 at 2025-07-22 19:12:33 UTC — 19 minutes after execution. The destination IP is a Cloudflare edge node fronting the attacker-controlled domain api-telegram-org.ctf.do. The malware uses Telegram Bot API (token: 8169267144:AAFwkHmXh71SMVcvFLantjTlwlRbT9HdJdU) as its C2 channel, abusing legitimate cloud infrastructure to blend with normal HTTPS traffic.

Impact: Active C2 channel established. Attacker has bidirectional communication with the compromised host. Telegram API provides file upload/download capabilities — data exfiltration or secondary payload delivery is highly probable. Cloudflare proxying makes IP-based blocking ineffective.

MITRE ATT&CK	TOOL	EVIDENCE	CROSS-VALIDATION
T1102.002, T1573	volatility3 2.7.0	EVID-001 · raw/vol3_netscan.csv	None — network capture unavailable for encrypted traffic analysis

Phishing Delivery Vector: Browser Download of Malicious Executable

Observation: epexplorer.exe was discovered at three virtual addresses in the memory filescan, including a partial browser download artifact (epexplorer.exe435568.crdownload). The .crdownload extension is specific to Chromium-based browsers (Microsoft Edge in this environment — confirmed by extensive Edge artifacts throughout the filescan). Per CTF intelligence, the phishing email originated from ktso.sec@mailforce.net.

Impact: User 'award' received a phishing email, clicked a link, and downloaded the malware via Microsoft Edge. The .crdownload artifact proves this was a browser-initiated download, not a drive-by or automated deployment. User awareness training gap identified. The phishing sender domain (mailforce.net) should be investigated for additional campaigns targeting the organization.

MITRE ATT&CK	TOOL	EVIDENCE	CROSS-VALIDATION
T1566.001, T1204.002	volatility3 2.7.0	EVID-001 · raw/ vol3_filesca.csv	None

Critical Asset Compromised: Air Traffic Control Workstation

Observation: The compromised host (TSO-ATC-CT412, user 'award') runs ATC_RADAR.exe — an air traffic control radar application with multiple active processes and network connections to 8.8.8.8 (Google DNS-over-HTTPS). The workstation name (ATC-CT412) and application context confirm this is not a standard enterprise endpoint — it is an aviation operations system. Multiple ATC_RADAR.exe instances ran concurrently with epexplorer.exe, meaning the malware coexisted with active ATC operations.

Impact: This is a worst-case compromise scenario. An attacker gained code execution on a safety-critical aviation system while it was actively processing radar data. Even if the malware's primary purpose was reconnaissance or C2 relay, the mere presence of unauthorized code on an ATC workstation constitutes a potential aviation safety incident. Regulatory reporting obligations may apply (ICAO Annex 17, national aviation authority).

MITRE ATT&CK	TOOL	EVIDENCE	CROSS-VALIDATION
T1003.001	volatility3 2.7.0 + MemProcFS 5.17.8	EVID-001 · raw/ vol3_pslist.csv	Dual-tool corroboration

Actionable IOCs

Blockable and searchable indicators extracted from findings. Use these to hunt for additional compromised hosts and block attacker infrastructure.

TYPE	INDICATOR	CONTEXT	SOURCE FINDING
SHA-256	3dc0d114859c0bde08d39155eaaa8f76392dd5121ca44ecb15652b3bf6049e35	Evidence archive	EVID-001
SHA-256	7b45e797acabaf592784bf7efceb979e3f036d5b9dd47f52280c17e38864c2dd	Memory dump	EVID-001
MD5	75ea94b54420c39dcd3d8ce574ba9d34	epxlorer.exe hash	F-niel-001
Filename	epxlorer.exe	Malware binary	F-niel-001
File Path	C:\Users\award\Downloads\epxlorer.exe	Execution path	F-niel-001
IP	104.21.1.247	C2 server	F-niel-002
Domain	api-telegram-org.ctf.do	C2 domain	F-niel-002
Email	ktso.sec@mailforce.net	Phishing sender	F-niel-003
IP	172.17.4.12	Compromised host	F-niel-004

Target System Identification

System metadata extracted from the memory dump. This is contextual information, not a finding.

PROPERTY	VALUE
Hostname	TSO-ATC-CT412
Username	award
Os	Windows 10 build 19041 (19041.1.amd64fre.vb_release.1912)
Architecture	x64
Processors	4
Host Ip	172.17.4.12
Dump Time	2025-07-22 19:18:53 UTC
Kernel Base	0xf80617e00000
Nt Product Type	NtProductWinNt
Source Tool	volatility3 2.7.0 windows.info.Info

Chain of Custody

All evidence items with cryptographic hashes and acquisition metadata.

EVIDENCE ID

EVID-001 – BelkaCTF_7_CASE250722_KTS0AERO.zip

SHA-256

3dc0d114859c0bde08d39155eaaa8f76392dd5121ca44ecb15652b3bf6049e35

SOURCE

https://dl.ctf.do/BelkaCTF_7_CASE250722_KTS0AERO.zip

Containment & Eradication

Prioritized remediation actions per finding. Execute in severity order (Critical → High → Medium).

F-NIEL-001

CRITICAL

Malware Execution: epexplorer.exe Masquerading as explorer.exe

Immediately isolate TSO-ATC-CT412 from all networks. Kill PID 9920. Quarantine C:\Users\award\Downloads\epexplorer.exe. Do NOT reboot before capturing full disk image — volatile evidence in memory will be lost.

F-NIEL-002

CRITICAL

Command and Control via Fraudulent Telegram Bot API

Block outbound HTTPS to api-telegram-org.ctf.do at network perimeter. Block Telegram Bot API endpoints if not business-justified. Deploy SSL inspection for traffic to unknown Cloudflare IPs. Check firewall logs for additional connections from 172.17.4.12 in the 18:50-19:18 UTC window.

F-NIEL-003

HIGH

Phishing Delivery Vector: Browser Download of Malicious Executable

Search mail server for emails from ktso.sec@mailforce.net — identify all recipients. Purge the phishing email from all inboxes. Block mailforce.net at email gateway. Initiate user awareness training for 'award' and ATC personnel. Review Edge browser security policies — consider disabling automatic execution of downloaded files.

F-NIEL-004

CRITICAL

Critical Asset Compromised: Air Traffic Control Workstation

Escalate to aviation safety officer and CISO immediately. This may require mandatory reporting to the national aviation authority. Ground the workstation — do not allow it to process live radar data until forensic imaging is complete and the system is rebuilt from trusted media. Engage the ATC software vendor (ATC_RADAR.exe) to verify software integrity. Review separation requirements between administrative workstations and operational ATC systems.

Limitations & Unresolved Questions

Areas not yet examined. Each gap represents uncertainty that should be addressed before final conclusions.

GAP 01 — DISK IMAGE NOT ANALYZED

Without filesystem forensics, persistence mechanisms and file-system timeline cannot be confirmed.

GAP 02 — NETWORK CAPTURE UNAVAILABLE

Encrypted C2 traffic cannot be analyzed for exfiltration volume.

GAP 03 — REGISTRY HIVES NOT EXAMINED

User activity artifacts and persistence locations remain unexamined.

GAP 04 — MALWARE NOT REVERSE-ENGINEERED

Full capability assessment of explorer.exe is pending.

GAP 05 — SUPPLEMENTARY EVIDENCE NOT ANALYZED

The 5.6 GB supplementary file was not analyzed.

GAP 06 — SINGLE HOST ANALYSIS ONLY

Compromise of additional systems cannot be ruled out.

Forensic Toolchain

Tools, versions, and runtimes used during this investigation. All tools validated by session canary before evidence analysis.

TOOL	VERSION	RUNTIME	VALIDATION STATUS
volatility3	2.7.0	docker	VALIDATED
memprocfs	5.17.8	host_fuse	VALIDATED
plaso	20240512	docker	VALIDATED
mft-tools	1.2.0.0	docker	VALIDATED
sift-vm	?	?	VALIDATED
session_canary	?	?	DEGRADED

Evidence Artifacts — Raw Tool Output

Unmodified tool output captured during analysis. Each artifact is traceable to the findings register and provides the raw data behind every conclusion in this report.

Evidence Artifacts — Raw Tool Output

Case: INC-2026-0624-0001 · 7 artifacts · 2026-06-24T22:24:48.372570

A1. filescanvol3_filescan.csv · 8,699 lines · 612 KB

Volatility 3

Framework 2.7.0

Offset	Name	Size
0x9f00000f5900	示***口示***口	0
0x9f00000f8510	\Windows\servicing\InboxFodMetadataCache\metadata\Language.Basic~bg-bg~1.0.mum	216
0x9f00000f86a0	\ProgramData\Microsoft\Windows\AppRepository\StateRepository-Machine.srd-wal	216
0x9f00000f8830	\Windows\System32\msdtc.exe	216
0x9f00000f89c0	\Windows\System32\CatRoot\{F750E6C3-38EE-11D1-85E5-00C04FC295EE}\Microsoft-Windows-Client-Desktop-Required-Package05113~31bf3856ad364e35~amd64~~10.0.19041.6093.cat	216
0x9f00000f8b50	\Windows\System32\Tasks\Microsoft\Windows\Windows Defender\Windows Defender Cache Maintenance	216
0x9f00000f8ce0	\Windows\System32\CatRoot\{F750E6C3-38EE-11D1-85E5-00C04FC295EE}\Microsoft-Windows-Client-Desktop-Required-Package051020~31bf3856ad364e35~amd64~~10.0.19041.6093.cat	216
0x9f00000f8e70	\Windows\servicing\InboxFodMetadataCache\metadata\Language.Basic~hu-hu~1.0.mum	216
0x9f00000f9190	\Program Files (x86)\Windows Live\Contacts\abssm.dll	216
0x9f00000f9320	\\$Directory	216
0x9f00000f9640	\\$Directory	216
0x9f00000f97d0	\Windows\System32\DdcComImplementationsDesktop.dll	216
0x9f00000f9960	\\$Directory	216
0x9f00000f9af0	\Program Files\WindowsApps\Microsoft.XboxIdentityProvider_12.50.6001.0_x64__8wekyb3d8bbwe\XboxIdp.exe	216
0x9f00000f9e10	\Program Files (x86)\Microsoft\EdgeUpdate\1.3.195.61\psmachine.dll	216
0x9f00001628f0	裂*裂*裂*	0
0x9f00001758e0	樓*樓*樓*	0
0x9f00001811b0	\Windows\System32\CatRoot\{F750E6C3-38EE-11D1-85E5-00C04FC295EE}\Microsoft-Windows-NetFx-Windows-Built-Package~31bf3856ad364e35~amd64~~10.0.19041.1.cat	216
0x9f0000181340	\Windows\SysWOW64\vm3dum-stats.dll	216
0x9f00001814d0	\\$Directory	216

0x9f0000181660	\\\$Directory	216
0x9f00001817f0	\\\$Directory	216
0x9f0000181980	\\\$Directory	216
0x9f0000181b10	\\Windows\\SysWOW64\\vm3dum_10-stats.dll	216
0x9f0000181ca0	\\\$Directory	216
0x9f0000181e30	\\Windows\\SysWOW64\\vmhgfs.dll	216
0xe689224b10c0	\\Windows\\System32\\dxgiadaptercache.exe	216
0xe68922afb0a0	\\Device\\HarddiskVolume3\\\$Extend\\\$RmMetadata\\\$TxfLog\\\$TxfLog	216
0xe68922afb380	\\Windows\\System32\\drivers\\hidparse.sys	216
0xe68922afb4f0	\\\$Secure:\$SDS:\$DATA	216
0xe68922afb660	\\\$Mft	216
0xe68922afb7d0	\\\$Extend\\\$RmMetadata\\\$Repair:\$Verify:\$DATA	216
0xe68922afbab0	\\\$MftMirr	216
0xe68922afbd90	\\\$Extend:\$I30:\$INDEX_ALLOCATION	216
0xe68922afc070	\\Windows\\System32\\smss.exe	216
0xe68922afc1e0	\\\$LogFile	216
0xe68922afc350	\\\$BitMap	216
0xe68922afc630	\\\$Mft::\$BITMAP	216
0xe68922afc7a0	\\:\$I30:\$INDEX_ALLOCATION	216
0xe68922afc910	\\\$Secure:\$SII:\$INDEX_ALLOCATION	216
0xe68922afca80	\\\$Extend\\\$RmMetadata\\\$Repair	216
0xe68922afcbf0	\\\$Extend\\\$Deleted:\$I30:\$INDEX_ALLOCATION	216
0xe68922afcd60	\\\$Extend\\\$Usnjrnl:\$J:\$DATA	216
0xe68922afced0	\\\$Extend\\\$RmMetadata\\\$Repair:\$Corrupt:\$DATA	216
0xe68922b28210	\\\$Extend\\\$RmMetadata\\\$TxfLog\\\$TxfLogContainer000000000000000000000001	216
0xe68922b28940	\\\$Directory	216
0xe68922b28c20	\\\$Extend\\\$RmMetadata\\\$TxfLog\\\$TxfLogContainer000000000000000000000002	216

Showing 50 of 8,698 rows

A2. hivelistvol3_hivelist.csv · 33 lines · 3 KB

Volatility 3

Framework 2.7.0

Offset	FileFullPath
0xaa8e0729a000	
0xaa8e07287000	\\REGISTRY\\MACHINE\\SYSTEM

0xaa8e07309000	\REGISTRY\MACHINE\HARDWARE
0xaa8e087fa000	\Device\HarddiskVolume1\EFI\Microsoft\Boot\BCD
0xaa8e087f8000	\SystemRoot\System32\Config\SOFTWARE
0xaa8e08991000	\SystemRoot\System32\Config\DEFAULT
0xaa8e08e20000	\SystemRoot\System32\Config\SECURITY
0xaa8e08e70000	\SystemRoot\System32\Config\SAM
0xaa8e08f89000	\\?\C:\Windows\ServiceProfiles\NetworkService\NTUSER.DAT
0xaa8e08d5f000	\SystemRoot\System32\Config\BB1
0xaa8e0b185000	\\?\C:\Windows\ServiceProfiles\LocalService\NTUSER.DAT
0xaa8e0c17d000	\\?\C:\Users\award\ntuser.dat
0xaa8e0c1d8000	\\?\C:\Users\award\AppData\Local\Microsoft\Windows\UsrClass.dat
0xaa8e0ce4e000	\\?\C:\Windows\AppCompat\Programs\Amcache.hve
0xaa8e0d3a6000	\\?\C: \ProgramData\Microsoft\Windows\AppRepository\Packages\Microsoft.Windows.StartMenuExperienceHost_10.0.19041.5438_ne
0xaa8e0d4bd000	\\?\C:\Users\award\AppData\Local\Packages\Microsoft.Windows.StartMenuExperienceHost_cw5n1h2txyewy\Settings\settings.d
0xaa8e0d6c4000	\\?\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\Microsoft.Windows.Search_1.14.18.19041_neutral_neutral_cw
0xaa8e0d6ed000	\\?\C:\Users\award\AppData\Local\Packages\Microsoft.Windows.Search_cw5n1h2txyewy\Settings\settings.dat
0xaa8e0dd2d000	\\?\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\MicrosoftWindows.Client.CBS_1000.19062.1000.0_x64__cw5n1
0xaa8e0d684000	\\?\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\Microsoft.SkypeApp_14.53.77.0_x64__kzf8qxf38zg5c\Activatio
0xaa8e0dfbb000	\\?\C:\Users\award\AppData\Local\Packages\Microsoft.SkypeApp_kzf8qxf38zg5c\Settings\settings.dat
0xaa8e0ede5000	\\?\C:\Users\award\AppData\Local\Packages\MicrosoftWindows.Client.CBS_cw5n1h2txyewy\Settings\settings.dat
0xaa8e185ce000	\\?\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\Microsoft.WindowsStore_11910.1002.5.0_x64__8wekyb3d8bbw
0xaa8e184e4000	\\?\C:\Users\award\AppData\Local\Packages\Microsoft.WindowsStore_8wekyb3d8bbwe\Settings\settings.dat
0xaa8e197b4000	\\?\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\windows.immersivecontrolpanel_10.0.2.1000_neutral_neutral
0xaa8e1988b000	\\?\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\Microsoft.Windows.ShellExperienceHost_10.0.19041.5072_neu
0xaa8e19c28000	\\?\C:\Users\award\AppData\Local\Packages\Microsoft.Windows.ShellExperienceHost_cw5n1h2txyewy\Settings\settings.dat
0xaa8e1b5d4000	\SystemRoot\System32\config\DRIVERS

A3. infovol3_info.txt · 139,720 lines · 9271 KB

Progress: 0.00 Updating caches for 66 files...

Progress: 1.52 Updating caches for 66 files...

Progress: 3.03 Updating caches for 66 files...

Progress: 4.55 Updating caches for 66 files...

Progress: 6.06 Updating caches for 66 files...

Progress:	7.58	Updating caches for 66 files...
Progress:	9.09	Updating caches for 66 files...
Progress:	10.61	Updating caches for 66 files...
Progress:	12.12	Updating caches for 66 files...
Progress:	13.64	Updating caches for 66 files...
Progress:	15.15	Updating caches for 66 files...
Progress:	16.67	Updating caches for 66 files...
Progress:	18.18	Updating caches for 66 files...
Progress:	19.70	Updating caches for 66 files...
Progress:	21.21	Updating caches for 66 files...
Progress:	22.73	Updating caches for 66 files...
Progress:	24.24	Updating caches for 66 files...
Progress:	25.76	Updating caches for 66 files...
Progress:	27.27	Updating caches for 66 files...
Progress:	28.79	Updating caches for 66 files...
Progress:	30.30	Updating caches for 66 files...
Progress:	31.82	Updating caches for 66 files...
Progress:	33.33	Updating caches for 66 files...
Progress:	34.85	Updating caches for 66 files...
Progress:	36.36	Updating caches for 66 files...
Progress:	37.88	Updating caches for 66 files...
Progress:	39.39	Updating caches for 66 files...
Progress:	40.91	Updating caches for 66 files...
Progress:	42.42	Updating caches for 66 files...

Progress:	43.94	Updating caches for 66 files...
Progress:	45.45	Updating caches for 66 files...
Progress:	46.97	Updating caches for 66 files...
Progress:	48.48	Updating caches for 66 files...
Progress:	50.00	Updating caches for 66 files...
Progress:	51.52	Updating caches for 66 files...
Progress:	53.03	Updating caches for 66 files...
Progress:	54.55	Updating caches for 66 files...
Progress:	56.06	Updating caches for 66 files...
Progress:	57.58	Updating caches for 66 files...
Progress:	59.09	Updating caches for 66 files...
Progress:	60.61	Updating caches for 66 files...
Progress:	62.12	Updating caches for 66 files...
Progress:	63.64	Updating caches for 66 files...
Progress:	65.15	Updating caches for 66 files...
Progress:	66.67	Updating caches for 66 files...
Progress:	68.18	Updating caches for 66 files...
Progress:	69.70	Updating caches for 66 files...
Progress:	71.21	Updating caches for 66 files...
Progress:	72.73	Updating caches for 66 files...
Progress:	74.24	Updating caches for 66 files...
Progress:	75.76	Updating caches for 66 files...
Progress:	77.27	Updating caches for 66 files...
Progress:	78.79	Updating caches for 66 files...

Progress:	80.30	Updating caches for 66 files...
Progress:	81.82	Updating caches for 66 files...
Progress:	83.33	Updating caches for 66 files...
Progress:	84.85	Updating caches for 66 files...
Progress:	86.36	Updating caches for 66 files...
Progress:	87.88	Updating caches for 66 files...
Progress:	89.39	Updating caches for 66 files...
Progress:	90.91	Updating caches for 66 files...
Progress:	92.42	Updating caches for 66 files...
Progress:	93.94	Updating caches for 66 files...
Progress:	95.45	Updating caches for 66 files...
Progress:	96.97	Updating caches for 66 files...
Progress:	98.48	Updating caches for 66 files...
Progress:	0.00	Scanning FileLayer using PageMapScanner
Progress:	23.33	Scanning FileLayer using PageMapScanner
Progress:	100.00	Stacking attempts finished
Progress:	0.00	Scanning memory_layer using BytesScanner
Progress:	0.31	Scanning memory_layer using BytesScanner
Progress:	0.63	Scanning memory_layer using BytesScanner
Progress:	0.00	Scanning layer_name using PdbSignatureScanner
Progress:	0.00	Scanning layer_name using PdbSignatureScanner
Progress:	0.10	Reading file http://msdl.microsoft.com/download/symbols/ntkrnl
Progress:	0.19	Reading file http://msdl.microsoft.com/download/symbols/ntkrnl
Progress:	0.29	Reading file http://msdl.microsoft.com/download/symbols/ntkrnl

Progress:	0.38	Reading file http://msdl.microsoft.com/download/symbols/ntkrn
Progress:	0.48	Reading file http://msdl.microsoft.com/download/symbols/ntkrn
Progress:	0.58	Reading file http://msdl.microsoft.com/download/symbols/ntkrn
Progress:	0.67	Reading file http://msdl.microsoft.com/download/symbols/ntkrn
Progress:	0.77	Reading file http://msdl.microsoft.com/download/symbols/ntkrn
Progress:	0.86	Reading file http://msdl.microsoft.com/download/symbols/ntkrn
Progress:	0.96	Reading file http://msdl.microsoft.com/download/symbols/ntkrn
Progress:	1.06	Reading file http://msdl.microsoft.com/download/symbols/ntkrn
Progress:	1.15	Reading file http://msdl.microsoft.com/download/symbols/ntkrn
Progress:	1.25	Reading file http://msdl.microsoft.com/download/symbols/ntkrn
Progress:	1.34	Reading file http://msdl.microsoft.com/download/symbols/ntkrn
Progress:	1.44	Reading file http://msdl.microsoft.com/download/symbols/ntkrn
Progress:	1.54	Reading file http://msdl.microsoft.com/download/symbols/ntkrn
Progress:	1.63	Reading file http://msdl.microsoft.com/download/symbols/ntkrn
Progress:	1.73	Reading file http://msdl.microsoft.com/download/symbols/ntkrn
Progress:	1.82	Reading file http://msdl.microsoft.com/download/symbols/ntkrn
Progress:	1.92	Reading file http://msdl.microsoft.com/download/symbols/ntkrn
Progress:	2.02	Reading file http://msdl.microsoft.com/download/symbols/ntkrn
Progress:	2.11	Reading file http://msdl.microsoft.com/download/symbols/ntkrn
Progress:	2.21	Reading file http://msdl.microsoft.com/download/symbols/ntkrn
Progress:	2.30	Reading file http://msdl.microsoft.com/download/symbols/ntkrn
Progress:	2.40	Reading file http://msdl.microsoft.com/download/symbols/ntkrn
Progress:	2.50	Reading file http://msdl.microsoft.com/download/symbols/ntkrn
Progress:	2.59	Reading file http://msdl.microsoft.com/download/symbols/ntkrn

Progress: 2.69

Reading file http://msdl.microsoft.com/download/symbols/ntkrn

Progress: 2.78

Reading file http://msdl.microsoft.com/download/symbols/ntkrn

Progress: 2.88

Reading file http://msdl.microsoft.com/download/symbols/ntkrn

Output truncated — full data in raw file

A4. malfindvol3_malfind.json · 275 lines · 16 KB

Volatility 3 Framework 2.7.0

PID	Process	Start VPN	End VPN	Tag	Protection	CommitCharge	PrivateMemory
3068	MsMpEng.exe	0x24b08a70000	0x24b08a70fff	VadS	PAGE_EXECUTE_READWRITE	1	
55	48 8d 2c 24 48 83 ec	UH.,\$H..					
20	48 8b 01 48 8b 49 08	.H..H.I.					
ff	d0 48 8d 65 00 5d c3	..H.e.]					
cc	cc cc cc cc cc cc cc cc						
cc	cc cc cc cc cc cc cc cc						
cc	cc cc cc cc cc cc cc cc						
cc	cc cc cc cc cc cc cc cc						
cc	cc cc cc cc cc cc cc cc		55 48 8d 2c 24 48 83 ec	20 48 8b 01 48 8b 49 08 ff d0			
3068	MsMpEng.exe	0x24b08a60000	0x24b08a60fff	VadS	PAGE_EXECUTE_READWRITE	1	
55	48 8d 2c 24 48 83 ec	UH.,\$H..					
20	48 8b 01 48 8b 49 08	.H..H.I.					
ff	d0 48 8d 65 00 5d c3	..H.e.]					
cc	cc cc cc cc cc cc cc cc						
cc	cc cc cc cc cc cc cc cc						
cc	cc cc cc cc cc cc cc cc						
cc	cc cc cc cc cc cc cc cc						
cc	cc cc cc cc cc cc cc cc		55 48 8d 2c 24 48 83 ec	20 48 8b 01 48 8b 49 08 ff d0			
3068	MsMpEng.exe	0x24b08ab0000	0x24b08ab1fff	VadS	PAGE_EXECUTE_READWRITE	2	
55	48 8d 2c 24 48 83 ec	UH.,\$H..					
20	48 8b 01 48 8b 49 08	.H..H.I.					
ff	d0 48 8d 65 00 5d c3	..H.e.]					
cc	cc cc cc cc cc cc cc cc						
cc	cc cc cc cc cc cc cc cc						
cc	cc cc cc cc cc cc cc cc						
cc	cc cc cc cc cc cc cc cc						
cc	cc cc cc cc cc cc cc cc		55 48 8d 2c 24 48 83 ec	20 48 8b 01 48 8b 49 08 ff d0			
3068	MsMpEng.exe	0x24b08b50000	0x24b08b50fff	VadS	PAGE_EXECUTE_READWRITE	1	
55	48 8d 2c 24 48 83 ec	UH.,\$H..					
20	48 8b 01 48 8b 49 08	.H..H.I.					
ff	d0 48 8d 65 00 5d c3	..H.e.]					
cc	cc cc cc cc cc cc cc cc						
cc	cc cc cc cc cc cc cc cc						
cc	cc cc cc cc cc cc cc cc						
cc	cc cc cc cc cc cc cc cc						
cc	cc cc cc cc cc cc cc cc		55 48 8d 2c 24 48 83 ec	20 48 8b 01 48 8b 49 08 ff d0			

3068	MsMpEng.exe	0x24b08b40000	0x24b08b42fff	VadS	PAGE_EXECUTE_READWRITE	3
55 48 8d 2c 24 48 83 ec	UH.,\$H..					
20 48 8b 01 48 8b 49 08	.H..H.I.					
ff d0 48 8d 65 00 5d c3	..H.e.]					
cc cc cc cc cc cc cc cc						
cc cc cc cc cc cc cc cc						
cc cc cc cc cc cc cc cc						
cc cc cc cc cc cc cc cc						
cc cc cc cc cc cc cc cc						
cc cc cc cc cc cc cc cc		55 48 8d 2c 24 48 83 ec	20 48 8b 01 48 8b 49 08	ff d0		
3068	MsMpEng.exe	0x24b08c40000	0x24b08c41fff	VadS	PAGE_EXECUTE_READWRITE	2
55 48 8d 2c 24 48 83 ec	UH.,\$H..					
20 48 8b 01 48 8b 49 08	.H..H.I.					
ff d0 48 8d 65 00 5d c3	..H.e.]					
cc cc cc cc cc cc cc cc						
cc cc cc cc cc cc cc cc						
cc cc cc cc cc cc cc cc						
cc cc cc cc cc cc cc cc						
cc cc cc cc cc cc cc cc						
cc cc cc cc cc cc cc cc		55 48 8d 2c 24 48 83 ec	20 48 8b 01 48 8b 49 08	ff d0		
3068	MsMpEng.exe	0x24b08ce0000	0x24b08ce0fff	VadS	PAGE_EXECUTE_READWRITE	1
55 48 8d 2c 24 48 83 ec	UH.,\$H..					
20 48 8b 01 48 8b 49 08	.H..H.I.					
ff d0 48 8d 65 00 5d c3	..H.e.]					
cc cc cc cc cc cc cc cc						
cc cc cc cc cc cc cc cc						
cc cc cc cc cc cc cc cc						
cc cc cc cc cc cc cc cc						
cc cc cc cc cc cc cc cc						
cc cc cc cc cc cc cc cc		55 48 8d 2c 24 48 83 ec	20 48 8b 01 48 8b 49 08	ff d0		
3068	MsMpEng.exe	0x24b08d20000	0x24b08d2bfff	VadS	PAGE_EXECUTE_READWRITE	12
55 48 8d 2c 24 48 83 ec	UH.,\$H..					
20 48 8b 01 48 8b 49 08	.H..H.I.					
ff d0 48 8d 65 00 5d c3	..H.e.]					
cc cc cc cc cc cc cc cc						
cc cc cc cc cc cc cc cc						
cc cc cc cc cc cc cc cc						
cc cc cc cc cc cc cc cc						
cc cc cc cc cc cc cc cc		55 48 8d 2c 24 48 83 ec	20 48 8b 01 48 8b 49 08	ff d0		
3068	MsMpEng.exe	0x24b0bb80000	0x24b0bc8cfff	VadS	PAGE_EXECUTE_READWRITE	269
56 57 53 55 41 54 41 55	VWSUATAU					
41 56 41 57 48 83 ec 28	AVAWH..(					
4c 8d 3c 24 48 8b e9 48	L.<\$H..H					
8d b1 98 38 00 00 ff e2	...8....					
49 8d 67 28 41 5f 41 5e	I.g(A_A^					
41 5d 41 5c 5d 5b 5f 5e	A]A\][_^					
c3 00 00 40 00 80 00 00	...@....					
00 48 89 e9 48 b8 80 61	.H..H..a	56 57 53 55 41 54 41 55	41 56 41 57 48 83 ec 28	4c 8d		
3068	MsMpEng.exe	0x24b0c680000	0x24b0c78cfff	VadS	PAGE_EXECUTE_READWRITE	269
56 57 53 55 41 54 41 55	VWSUATAU					
41 56 41 57 48 83 ec 28	AVAWH..(					

0x9f00001eca20	TCPv4	172.17.4.12	50286	8.8.8.8	443	ESTABLISHED	9456	ATC_RADAR.exe	2025-07-22 19:15:10.000
0x9f00001fe2e0	TCPv4	172.17.4.12	50287	8.8.8.8	443	ESTABLISHED	9456	ATC_RADAR.exe	2025-07-22 19:15:10.000
0xe68922caca70	TCPv4	0.0.0.0	49668	0.0.0.0	0	LISTENING	2540	spoolsv.exe	2025-07-22 18:50:48.000
0xe68922e59010	TCPv4	172.17.4.12	49871	18.165.122.73	443	ESTABLISHED	2256	msedge.exe	2025-07-22 18:51:58.000
0xe68922e5d010	TCPv4	172.17.4.12	50283	2.23.227.215	443	CLOSED	2256	msedge.exe	2025-07-22 19:15:09.000
0xe689238948a0	TCPv4	172.17.4.12	50241	104.21.1.247	443	ESTABLISHED	9920	explorer.exe	2025-07-22 19:12:33.000
0xe68923afc050	TCPv4	0.0.0.0	49664	0.0.0.0	0	LISTENING	700	lsass.exe	2025-07-22 18:50:48.000
0xe68923afc050	TCPv6	::	49664	::	0	LISTENING	700	lsass.exe	2025-07-22 18:50:48.000
0xe68923afc310	TCPv4	0.0.0.0	49665	0.0.0.0	0	LISTENING	536	wininit.exe	2025-07-22 18:50:48.000
0xe68923afc470	TCPv4	0.0.0.0	49668	0.0.0.0	0	LISTENING	2540	spoolsv.exe	2025-07-22 18:50:48.000
0xe68923afc470	TCPv6	::	49668	::	0	LISTENING	2540	spoolsv.exe	2025-07-22 18:50:48.000
0xe68923afc730	TCPv4	0.0.0.0	49667	0.0.0.0	0	LISTENING	1332	svchost.exe	2025-07-22 18:50:48.000
0xe68923afc9f0	TCPv4	0.0.0.0	135	0.0.0.0	0	LISTENING	940	svchost.exe	2025-07-22 18:50:48.000
0xe68923afcb50	TCPv4	0.0.0.0	135	0.0.0.0	0	LISTENING	940	svchost.exe	2025-07-22 18:50:48.000
0xe68923afcb50	TCPv6	::	135	::	0	LISTENING	940	svchost.exe	2025-07-22 18:50:48.000
0xe68923afccb0	TCPv4	0.0.0.0	49665	0.0.0.0	0	LISTENING	536	wininit.exe	2025-07-22 18:50:48.000
0xe68923afccb0	TCPv6	::	49665	::	0	LISTENING	536	wininit.exe	2025-07-22 18:50:48.000
0xe68923afd4f0	TCPv4	0.0.0.0	49667	0.0.0.0	0	LISTENING	1332	svchost.exe	2025-07-22 18:50:48.000
0xe68923afd4f0	TCPv6	::	49667	::	0	LISTENING	1332	svchost.exe	2025-07-22 18:50:48.000
0xe68923afd7b0	TCPv4	172.17.4.12	139	0.0.0.0	0	LISTENING	4	System	2025-07-22 18:50:50.000
0xe68923afd910	TCPv4	0.0.0.0	49666	0.0.0.0	0	LISTENING	1032	svchost.exe	2025-07-22 18:50:48.000
0xe68923afd910	TCPv6	::	49666	::	0	LISTENING	1032	svchost.exe	

									2025-07-22 18:50:48.000
0xe68923afda70	TCPv4	0.0.0.0	49666	0.0.0.0	0	LISTENING	1032	svchost.exe	2025-07-22 18:50:48.000
0xe68923afdbd0	TCPv4	0.0.0.0	49664	0.0.0.0	0	LISTENING	700	lsass.exe	2025-07-22 18:50:48.000
0xe68923afdd30	TCPv4	0.0.0.0	445	0.0.0.0	0	LISTENING	4	System	2025-07-22 18:50:49.000
0xe68923afdd30	TCPv6	::	445	::	0	LISTENING	4	System	2025-07-22 18:50:49.000
0xe689244768a0	TCPv4	172.17.4.12	49869	20.93.72.182	443	ESTABLISHED	2256	msedge.exe	2025-07-22 18:51:56.000
0xe6892459a050	TCPv4	172.17.4.12	50244	2.23.227.215	443	ESTABLISHED	7012	SearchApp.exe	2025-07-22 19:14:19.000
0xe68924f62730	TCPv4	0.0.0.0	9999	0.0.0.0	0	LISTENING	6892	ATC_RADAR.exe	2025-07-22 19:15:06.000
0xe68924f62cb0	TCPv4	0.0.0.0	5040	0.0.0.0	0	LISTENING	5084	svchost.exe	2025-07-22 18:50:55.000
0xe689275e1050	TCPv4	172.17.4.12	49924	48.209.180.244	443	ESTABLISHED	7888	smartscreen.ex	2025-07-22 18:53:08.000
0xe689276a0140	UDPv4	127.0.0.1	49663	*	0		3144	svchost.exe	2025-07-22 18:50:49.000
0xe689276fd470	TCPv4	0.0.0.0	49669	0.0.0.0	0	LISTENING	680	services.exe	2025-07-22 18:50:49.000
0xe689276fd470	TCPv6	::	49669	::	0	LISTENING	680	services.exe	2025-07-22 18:50:49.000
0xe689276fd5d0	TCPv4	0.0.0.0	49669	0.0.0.0	0	LISTENING	680	services.exe	2025-07-22 18:50:49.000
0xe68927895c00	UDPv4	0.0.0.0	0	*	0		3924	svchost.exe	2025-07-22 18:50:49.000
0xe68927895d90	UDPv4	0.0.0.0	0	*	0		3924	svchost.exe	2025-07-22 18:50:49.000
0xe68927895d90	UDPv6	::	0	*	0		3924	svchost.exe	2025-07-22 18:50:49.000
0xe68927b39b00	TCPv4	172.17.4.12	50295	135.234.160.246	443	CLOSED	2896	wermgr.exe	2025-07-22 19:15:14.000
0xe68927d768a0	TCPv4	172.17.4.12	50294	20.190.147.11	443	CLOSED	6048	svchost.exe	2025-07-22 19:15:13.000
0xe68927d7fb00	TCPv4	172.17.4.12	50259	13.107.246.254	443	ESTABLISHED	7012	SearchApp.exe	2025-07-22 19:14:22.000
0xe68927eb66e0	UDPv4	0.0.0.0	5355	*	0		2168	svchost.exe	2025-07-22 18:50:50.000
0xe68927eb66e0	UDPv6	::	5355	*	0		2168	svchost.exe	2025-07-22 18:50:50.000

0xe68927eb7360	UDPv4	172.17.4.12	138	*	0	4	System	2025-07-22 18:50:50.000
0xe68927eb7680	UDPv4	0.0.0.0	0	*	0	2168	svchost.exe	2025-07-22 18:50:50.000
0xe68927eb7680	UDPv6	::	0	*	0	2168	svchost.exe	2025-07-22 18:50:50.000
0xe68927eb7cc0	UDPv4	0.0.0.0	5353	*	0	2168	svchost.exe	2025-07-22 18:50:50.000

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A6. pslistvol3_pslist.csv · 154 lines · 13 KB

Volatility 3 Framework 2.7.0

PID	PPID	ImageFileName	Offset(V)	Threads	Handles	SessionId	Wow64	CreateTime	ExitTime	PID
4	0	System	0xe68922462040	133	-	N/A	False	2025-07-22 18:50:46.000000	N/A	D
108	4	Registry	0xe689224da080	4	-	N/A	False	2025-07-22 18:50:41.000000	N/A	D
344	4	smss.exe	0xe68922cdc040	2	-	N/A	False	2025-07-22 18:50:46.000000	N/A	D
460	448	csrss.exe	0xe689231f6080	10	-	0	False	2025-07-22 18:50:47.000000	N/A	D
536	448	wininit.exe	0xe68923aaf0c0	1	-	0	False	2025-07-22 18:50:47.000000	N/A	D
544	528	csrss.exe	0xe68923b20140	13	-	1	False	2025-07-22 18:50:47.000000	N/A	D
636	528	winlogon.exe	0xe68923b95080	5	-	1	False	2025-07-22 18:50:48.000000	N/A	D
680	536	services.exe	0xe68922b4a080	5	-	0	False	2025-07-22 18:50:48.000000	N/A	D
700	536	lsass.exe	0xe68923a77280	10	-	0	False	2025-07-22 18:50:48.000000	N/A	D
820	680	svchost.exe	0xe68924246280	13	-	0	False	2025-07-22 18:50:48.000000	N/A	D
844	636	fontdrvhost.ex	0xe689242440c0	5	-	1	False	2025-07-22 18:50:48.000000	N/A	D
852	536	fontdrvhost.ex	0xe68924248180	5	-	0	False	2025-07-22 18:50:48.000000	N/A	D
940	680	svchost.exe	0xe689242be300	8	-	0	False	2025-07-22 18:50:48.000000	N/A	D
996	680	svchost.exe	0xe689242bc0c0	5	-	0	False	2025-07-22 18:50:48.000000	N/A	D

756	680	svchost.exe	0xe68924393340	3	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1032	680	svchost.exe	0xe689243b2340	7	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1056	680	svchost.exe	0xe689243c12c0	5	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1064	680	svchost.exe	0xe689243bf0c0	3	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1140	636	dwm.exe	0xe689243da080	13	-	1	False	2025-07-22 18:50:48.000000	N/A	D
1168	680	svchost.exe	0xe689243de300	1	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1256	680	svchost.exe	0xe6892442c340	5	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1332	680	svchost.exe	0xe68924473280	7	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1396	680	svchost.exe	0xe689244bb340	5	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1420	680	svchost.exe	0xe689244bd2c0	2	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1440	680	svchost.exe	0xe689244d7340	2	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1596	680	svchost.exe	0xe689245762c0	3	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1616	680	svchost.exe	0xe68924510340	4	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1652	680	svchost.exe	0xe689245b02c0	2	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1664	680	svchost.exe	0xe689245b42c0	2	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1772	680	svchost.exe	0xe68924626340	7	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1820	680	svchost.exe	0xe6892463c340	1	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1848	4	MemCompression	0xe68924642040	42	-	N/A	False	2025-07-22 18:50:48.000000	N/A	D
1860	680	svchost.exe	0xe68924644280	2	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1960	1332	taskhostw.exe	0xe689245b2080	0	-	0	False	2025-07-22 18:50:48.000000	2025-07-22 18:51:23.000000	D
1972	680	svchost.exe	0xe689245b1080	2	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1980	680	svchost.exe	0xe689246c3340	5	-	0	False		N/A	D

								2025-07-22 18:50:48.000000		
2000	680	svchost.exe	0xe689246d6300	2	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1868	680	svchost.exe	0xe68924724080	5	-	0	False	2025-07-22 18:50:48.000000	N/A	D
2168	680	svchost.exe	0xe6892721d340	10	-	0	False	2025-07-22 18:50:48.000000	N/A	D
2180	680	svchost.exe	0xe689247350c0	3	-	0	False	2025-07-22 18:50:48.000000	N/A	D
2196	680	svchost.exe	0xe689247f0340	4	-	0	False	2025-07-22 18:50:48.000000	N/A	D
2272	680	svchost.exe	0xe689272822c0	2	-	0	False	2025-07-22 18:50:48.000000	N/A	D
2372	680	svchost.exe	0xe689273020c0	5	-	0	False	2025-07-22 18:50:48.000000	N/A	D
2540	680	spoolsv.exe	0xe68922520080	7	-	0	False	2025-07-22 18:50:48.000000	N/A	D
2600	680	svchost.exe	0xe689224ee080	13	-	0	False	2025-07-22 18:50:48.000000	N/A	D
2660	680	svchost.exe	0xe689225ca080	5	-	0	False	2025-07-22 18:50:48.000000	N/A	D
2880	680	svchost.exe	0xe689273f2280	22	-	0	False	2025-07-22 18:50:49.000000	N/A	D

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A7. windows psscan PsScanvol3_windows_psscan_PsScan.csv · 167 lines · 15 KB

Volatility 3
Framework
2.7.0

PID	PPID	ImageFileName	Offset(V)	Threads	Handles	SessionId	Wow64	CreateTime	ExitTime	F
6224	1332	taskhostw.exe	0x9f000006f300	3	-	1	False	2025-07-22 18:50:52.000000	N/A	D
4	0	System	0xe68922462040	133	-	N/A	False	2025-07-22 18:50:46.000000	N/A	D
108	4	Registry	0xe689224da080	4	-	N/A	False	2025-07-22 18:50:41.000000	N/A	D
2600	680	svchost.exe	0xe689224ee080	13	-	0	False	2025-07-22 18:50:48.000000	N/A	D
2540	680	spoolsv.exe	0xe68922520080	7	-	0	False	2025-07-22 18:50:48.000000	N/A	D
2660	680	svchost.exe	0xe689225ca080	5	-	0	False	2025-07-22 18:50:48.000000	N/A	D

680	536	services.exe	0xe68922b4a080	5	-	0	False	2025-07-22 18:50:48.000000	N/A	D
344	4	smss.exe	0xe68922cdc040	2	-	N/A	False	2025-07-22 18:50:46.000000	N/A	D
460	448	csrss.exe	0xe689231f6080	10	-	0	False	2025-07-22 18:50:47.000000	N/A	D
8524	680	svchost.exe	0xe689238fa080	3	-	0	False	2025-07-22 18:56:49.000000	N/A	D
700	536	lsass.exe	0xe68923a77280	10	-	0	False	2025-07-22 18:50:48.000000	N/A	D
536	448	wininit.exe	0xe68923aaf0c0	1	-	0	False	2025-07-22 18:50:47.000000	N/A	D
544	528	csrss.exe	0xe68923b20140	13	-	1	False	2025-07-22 18:50:47.000000	N/A	D
1608	8180	ATC_RADAR.exe	0xe68923b6a080	0	-	1	False	2025-07-22 19:08:53.000000	2025-07-22 19:15:02.000000	D
636	528	winlogon.exe	0xe68923b95080	5	-	1	False	2025-07-22 18:50:48.000000	N/A	D
844	636	fontdrvhost.ex	0xe689242440c0	5	-	1	False	2025-07-22 18:50:48.000000	N/A	D
820	680	svchost.exe	0xe68924246280	13	-	0	False	2025-07-22 18:50:48.000000	N/A	D
852	536	fontdrvhost.ex	0xe68924248180	5	-	0	False	2025-07-22 18:50:48.000000	N/A	D
996	680	svchost.exe	0xe689242bc0c0	5	-	0	False	2025-07-22 18:50:48.000000	N/A	D
940	680	svchost.exe	0xe689242be300	8	-	0	False	2025-07-22 18:50:48.000000	N/A	D
756	680	svchost.exe	0xe68924393340	3	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1032	680	svchost.exe	0xe689243b2340	7	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1064	680	svchost.exe	0xe689243bf0c0	3	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1056	680	svchost.exe	0xe689243c12c0	5	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1140	636	dwm.exe	0xe689243da080	13	-	1	False	2025-07-22 18:50:48.000000	N/A	D
1168	680	svchost.exe	0xe689243de300	1	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1256	680	svchost.exe	0xe6892442c340	5	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1332	680	svchost.exe	0xe68924473280	7	-	0	False		N/A	D

								2025-07-22 18:50:48.000000		
1396	680	svchost.exe	0xe689244bb340	5	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1420	680	svchost.exe	0xe689244bd2c0	2	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1440	680	svchost.exe	0xe689244d7340	2	-	0	False	2025-07-22 18:50:48.000000	N/A	D
4628	8180	ATC_RADAR.exe	0xe6892450d080	0	-	1	False	2025-07-22 19:08:53.000000	2025-07-22 19:15:02.000000	D
1616	680	svchost.exe	0xe68924510340	4	-	0	False	2025-07-22 18:50:48.000000	N/A	D
10096	680	svchost.exe	0xe689245452c0	1	-	0	False	2025-07-22 18:51:29.000000	N/A	D
10928	820	SystemSettings	0xe68924568080	17	-	1	False	2025-07-22 18:51:59.000000	N/A	D
1596	680	svchost.exe	0xe689245762c0	3	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1652	680	svchost.exe	0xe689245b02c0	2	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1972	680	svchost.exe	0xe689245b1080	2	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1960	1332	taskhostw.exe	0xe689245b2080	0	-	0	False	2025-07-22 18:50:48.000000	2025-07-22 18:51:23.000000	D
1664	680	svchost.exe	0xe689245b42c0	2	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1772	680	svchost.exe	0xe68924626340	7	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1820	680	svchost.exe	0xe6892463c340	1	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1848	4	MemCompression	0xe68924642040	42	-	N/A	False	2025-07-22 18:50:48.000000	N/A	D
1860	680	svchost.exe	0xe68924644280	2	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1980	680	svchost.exe	0xe689246c3340	5	-	0	False	2025-07-22 18:50:48.000000	N/A	D
2000	680	svchost.exe	0xe689246d6300	2	-	0	False	2025-07-22 18:50:48.000000	N/A	D
1868	680	svchost.exe	0xe68924724080	5	-	0	False	2025-07-22 18:50:48.000000	N/A	D

EXAMINER SIGNATURE

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This report is auto-generated from structured case data. All findings require human review before finalization.